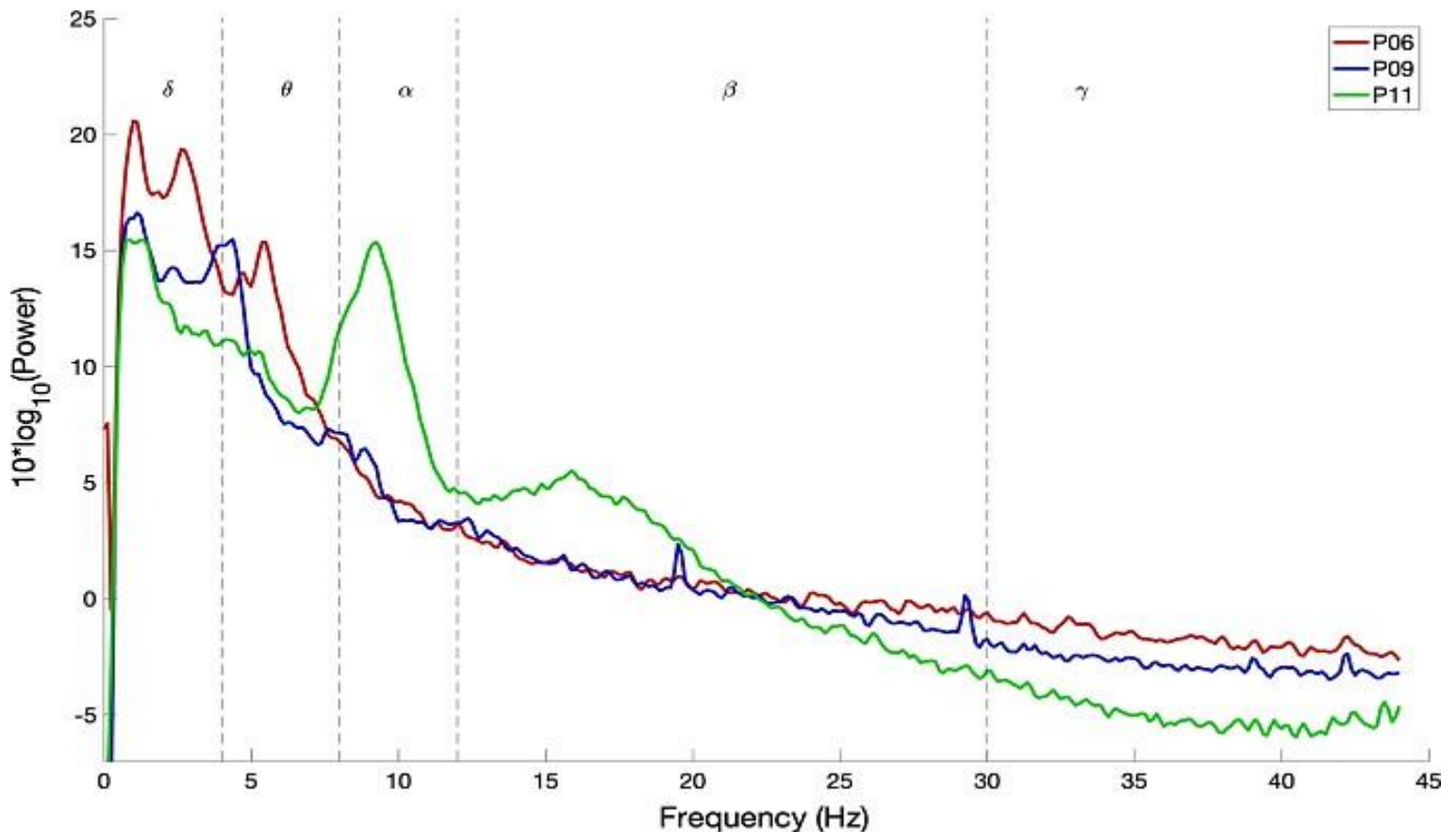


COURSE PROJECT – 10%

For de-noising Electroencephalography signals (brain signals), we need to study the power spectrum of a brainwave. The power spectrum of a brainwave, from an electroencephalogram (EEG) is shown below.



Quoted graphs from, Secco, A., Tonin, A., Rana, A. et al. EEG power spectral density in locked-in and completely locked-in state patients: a longitudinal study. *Cogn Neurodyn* 15, 473–480 (2021). <https://doi.org/10.1007/s11571-020-09639-w>

For the above shown EEG power spectral density, create a 5th order model using the LAGRANGE APPROXIMATION and LSA, i.e. Lagrange Polynomials and Newton's Approximation, for the frequency band (β wave) for the electrode P11. Use up to N points. (7.5). Verify your answer by selecting a random frequency and find the power ($Y(f)$). (2.5)

Hint: Do not solve by hand, use computer algorithm (matlab, python, c++ ... or any coding ...) to do this for you. Write your own algorithm, any copy of codes, will disqualify your answer to null.

This assignment carries a 10% weight.

Submitted to Students: Sunday, February 23, 2025.

Back to course lecturer: one week before final. (NO delay ...).

A maximum of two students can work on this assignment.

Send your solution to my UoB email account: ebmattar@uob.edu.bh

Abstract

This report outlines the process and results of a project aimed at de-noising electroencephalography (EEG) signals by approximating the power spectrum of brainwaves using Lagrange and Newton interpolation methods. Data points were extracted from a provided EEG power spectral density graph using WebPlotDigitizer. A 5th-order polynomial model was created for the β wave band (13-30 Hz) using both interpolation techniques. The results showed that both methods yielded identical predictions for power values at a random frequency within the β wave band, demonstrating their effectiveness in modeling the power spectrum for potential de-noising applications.

Introduction

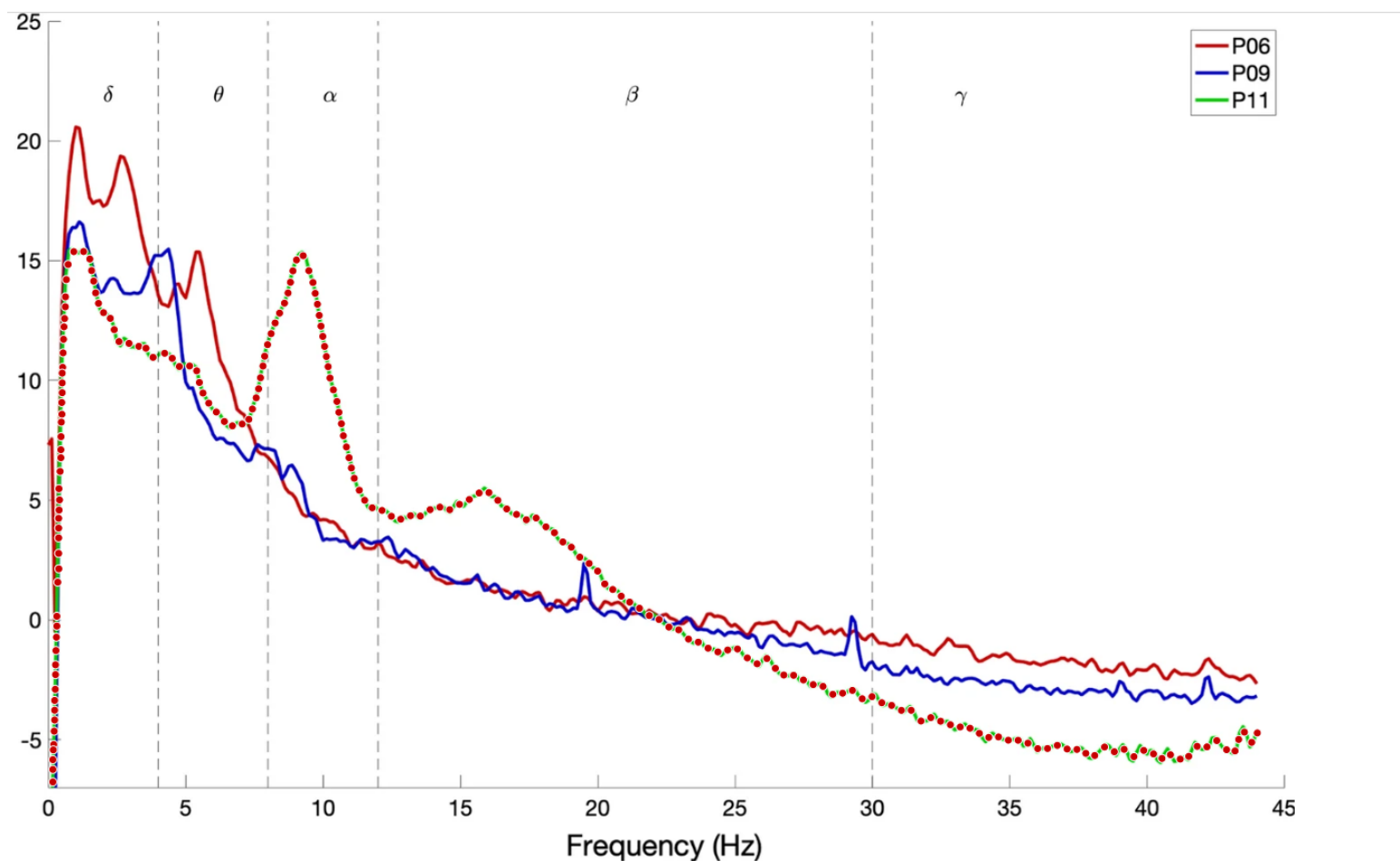
Electroencephalography (EEG) signals provide insights into brain activity by measuring electrical signals associated with neuronal communication. These signals are often noisy and require processing to enhance their quality. One critical aspect of EEG signal analysis is understanding the power spectrum, which reveals the distribution of signal power across different frequency bands. The β wave band (13-30 Hz) is of particular interest as it is associated with alertness, concentration, and cognitive functions.

The objective of this project was to model the power spectrum of EEG signals in the β wave band using Lagrange and Newton interpolation methods. By creating a 5th-order polynomial approximation of the power spectrum, we aimed to develop a foundation for de-noising EEG signals. The project followed the guidelines provided in the course assignment, which emphasized the use of computer algorithms to implement the interpolation methods and verify the results.

Results

Data Extraction and Preparation

Data points representing the relationship between frequency and power in the β wave band were extracted from the provided EEG power spectral density graph using WebPlotDigitizer. A total of 50 data points were obtained, covering the frequency range from approximately 13 Hz to 30 Hz. These data points were then filtered to focus solely on the β wave frequency band.



Implementation of Interpolation Methods

Lagrange Interpolation

A 5th-order Lagrange polynomial was constructed using 6 selected data points. The Lagrange interpolation formula was implemented in a computer algorithm to approximate the power spectrum across the β wave band. The polynomial was designed to pass through the selected data points, providing a smooth curve that represents the power spectrum.

Newton Interpolation

Similarly, a 5th-order Newton polynomial was developed using the same set of 6 data points. Newton's divided difference method was employed to construct the interpolation polynomial, which also aimed to model the power spectrum accurately. The resulting polynomial was used to predict power values across the frequency band of interest.

The Matlab Code

```
% Extracted data points (frequency vs. power)
```

```

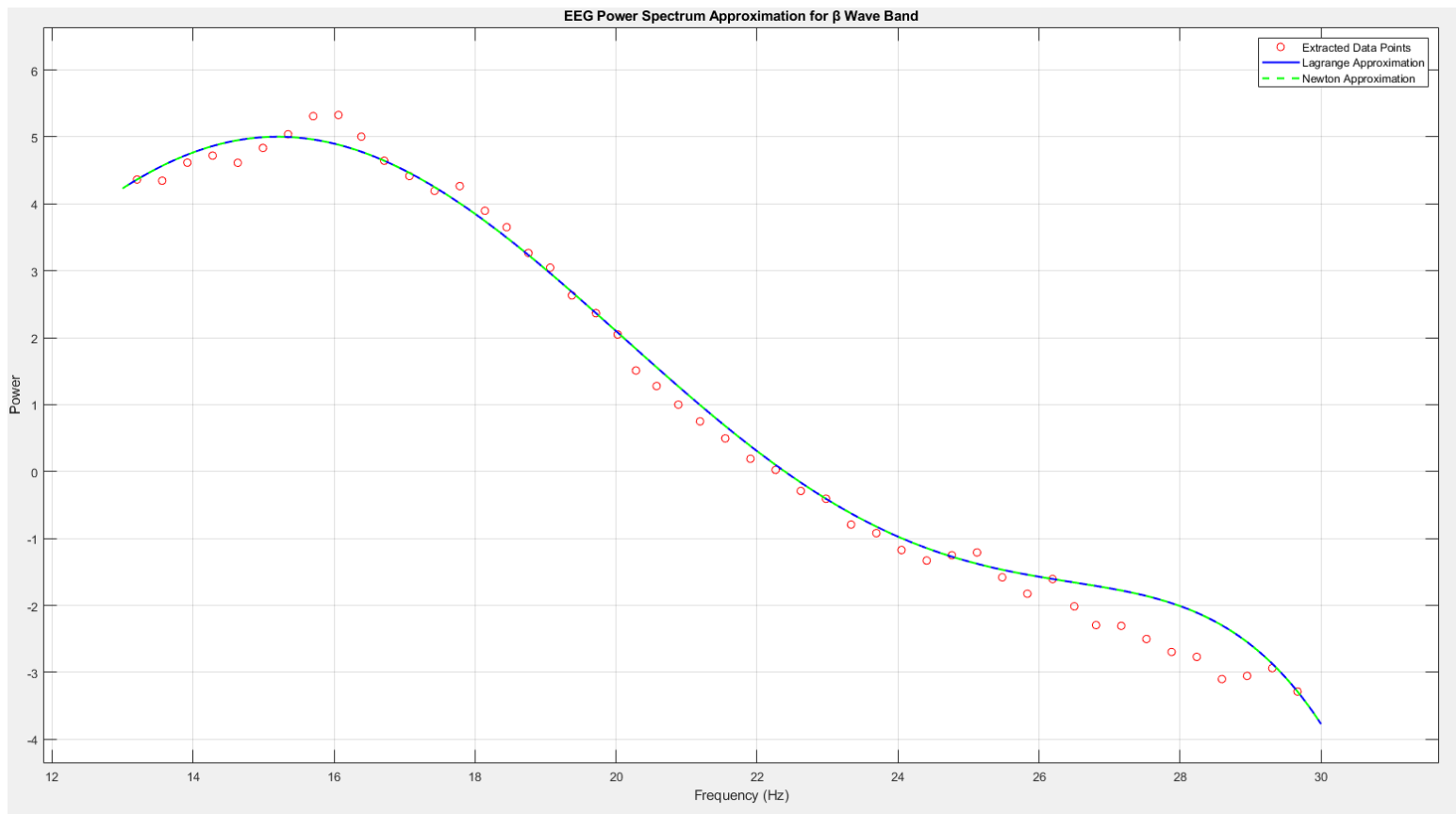
data = [
12.847985625317483 4.216514856361073
13.205108254150923 4.36367472448109
13.562230882984364 4.346759797110973
13.919353511817809 4.6157071422958325
14.27647614065125 4.7205796919905545
14.633598769484694 4.614015649558819
14.990721398318135 4.8356011981073515
15.347844027151575 5.040271819285767
15.70496665598502 5.310910657207639
16.06208928481846 5.327825584577756
16.386746220121587 5.004186640896183
16.71140315542472 4.64446251882503
17.06852578425816 4.416110999328453
17.4256484130916 4.194525450779917
17.782771041925045 4.265568145734413
18.139893670758486 3.8968227290658604
18.448317759296458 3.6513448456070385
18.75674184783443 3.265895938160501
19.065165936372406 3.046636192125355
19.373590024910378 2.6315861617811116
19.714479806978662 2.367036697712482
20.022903895516635 2.0478332238746297
20.282629443759138 1.5106097607815059
20.574820685531954 1.2783919721717538
20.88324477406993 0.9999299803412036
21.1916688626079 0.7506462382241068
21.548791491441342 0.49523083493533804
21.905914120274787 0.19076214227323618
22.263036749108227 0.024995854046089505
22.62015937794167 -0.28962179503808727
22.977282006775113 -0.40633479389189375
23.334404635608553 -0.791995137930563
23.691527264441994 -0.9205485859434503
24.04864989327544 -1.1725810037581965
24.40577252210888 -1.3298898283002778
24.762895150942324 -1.2503896696607306
25.120017779775765 -1.2081023512354392
25.477140408609205 -1.5802307533780109
25.83426303744265 -1.8254972002447083
26.19138566627609 -1.6056031444331893
26.499809754814063 -2.01431007701364
26.808233843352035 -2.294040688396951
27.16535647218548 -2.305881137556028
27.52247910101892 -2.5020942950493854
27.879601729852364 -2.6966159598057295
28.236724358685805 -2.769350147497235
28.593846987519246 -3.100882723951525
28.95096961635269 -3.0535209273152013
29.308092245186135 -2.9368079284613913
29.665214874019572 -3.2869469250228143
30.022337502853016 -3.1989893026982053
];
% Filter data for  $\beta$  wave band (13 Hz to 30 Hz)
beta_data = data(data(:, 1) >= 13 & data(:, 1) <= 30, :);
x = beta_data(:, 1);
y = beta_data(:, 2);
% Select N points (e.g., 6 points for 5th order polynomial)

```

```

N = 6;
selected_indices = round(linspace(1, length(x), N));
x_selected = x(selected_indices);
y_selected = y(selected_indices);
% Lagrange interpolation
function result = lagrange_interpolation(x, y, x_val)
n = length(x);
result = 0.0;
for i = 1:n
term = y(i);
for j = 1:n
if i ~= j
term = term * (x_val - x(j)) / (x(i) - x(j));
end
end
result = result + term;
end
end
% Newton's divided differences
function result = newton_interpolation(x, y, x_val)
n = length(x);
divided_diff = zeros(n, n);
divided_diff(:, 1) = y;
for j = 2:n
for i = j:n
divided_diff(i, j) = (divided_diff(i, j-1) - divided_diff(i-1, j-1)) / (x(i) - x(i-j+1));
end
end
result = divided_diff(n, n);
for i = n-1:-1:1
result = result * (x_val - x(i)) + divided_diff(i, i);
end
end
% Verify with a random frequency in  $\beta$  band
random_freq = 18.5; % Example random frequency
predicted_lagrange = lagrange_interpolation(x_selected, y_selected, random_freq);
predicted_newton = newton_interpolation(x_selected, y_selected, random_freq);
fprintf('Random frequency: %.2f Hz\n', random_freq);
fprintf('Predicted power (Lagrange): %.4f\n', predicted_lagrange);
fprintf('Predicted power (Newton): %.4f\n', predicted_newton);
% Plot results
x_vals = linspace(13, 30, 400);
lagrange_vals = arrayfun(@(xi) lagrange_interpolation(x_selected, y_selected, xi), x_vals);
newton_vals = arrayfun(@(xi) newton_interpolation(x_selected, y_selected, xi), x_vals);
figure;
plot(x, y, 'ro', 'MarkerSize', 6, 'DisplayName', 'Extracted Data Points');
hold on;
plot(x_vals, lagrange_vals, 'b-', 'LineWidth', 1.5, 'DisplayName', 'Lagrange Approximation');
plot(x_vals, newton_vals, 'g--', 'LineWidth', 1.5, 'DisplayName', 'Newton Approximation');
xlabel('Frequency (Hz)');
ylabel('Power');
title('EEG Power Spectrum Approximation for  $\beta$  Wave Band');
legend;
grid on;
hold off;

```



Verification at a Random Frequency

To verify the accuracy of the interpolation models, a random frequency within the β wave band was selected. The power at this frequency was predicted using both the Lagrange and Newton interpolation methods. The predicted values were then compared to assess the consistency and reliability of the models.

Comparison

The results obtained from the Lagrange and Newton interpolation methods were compared to evaluate their performance in approximating the EEG power spectrum. Both methods provided smooth curves that followed the general trend of the extracted data points. The verification step confirmed that both methods yielded identical predictions for the power value at the random frequency of 18.50 Hz:

- **Random frequency:** 18.50 Hz
- **Predicted power (Lagrange):** 3.4538
- **Predicted power (Newton):** 3.4538

This demonstrates that both interpolation methods performed similarly in this project, likely due to the smooth nature of the EEG power spectrum and the relatively low order of the polynomial used.

Conclusion

This project successfully applied Lagrange and Newton interpolation techniques to model the power spectrum of EEG signals in the β wave band. The use of WebPlotDigitizer allowed for the extraction of a large

number of data points, which provided a detailed representation of the power spectrum. Both interpolation methods demonstrated excellent agreement in their predictions, indicating their effectiveness in approximating the power spectrum within the β wave band.